

Roll No.

Total Pages : 06

013303

December 2024

B.Tech. (ME) (Third Semester)

Fluid Mechanics and Machines

(PCC-ME-303/21)

Time : 3 Hours]

[Maximum Marks : 75

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other. Assume suitable missing data.

Part A

1. (a) Define specific gravity of a liquid and a gas. 1.5
- (b) Write the stability conditions of a floating body. 1.5
- (c) Write impulse momentum relation. Write any one application of momentum equation. 1.5

(d) Define momentum thickness of a boundary layer. 1.5

(e) Write two formulas to determine head loss due to friction in a flow through a pipe. 1.5

(f) Define similitude. What are different types of similitude ? 1.5

(g) What are the advantages of using a draft tube in a reaction turbine ? 1.5

(h) Draw a neat sketch of a Kaplan turbine. 1.5

(i) Define manometric head of a centrifugal pump. 1.5

(j) Draw the ideal indicator diagram for a single acting reciprocating pumps ? 1.5

Part B

2. (a) Classify different types of fluids. 7
- (b) Find out the shear stress created by a fluid having dynamic viscosity of 10 poise and flowing over a flat plate at a distance of 0.2 m. If the velocity of the fluid is represented by v (m/sec) at a distance of y (m) from the plate. The velocity distribution is given as $v = 0.5y - y^2$ at a distance in which u is the velocity in metre per second at a distance y metre above the plate. 8
3. (a) Derive a general form of continuity equation. 7
- (b) Determine the resultant force experienced by a right-angle bend fitted with a pipe having a diameter of 250 mm in horizontal plane. The total discharge flowing through the pipe and bend is $0.250 \text{ m}^3/\text{sec}$. Consider the pressures at the inlet and outlet of the bend as 24 N/cm^2 and 23 N/cm^2 , respectively. 8

4. (a) Derive Hagen-Poiseuille equation for laminar flow in the circular pipe. 7

(b) A laminar flow of oil occurs between two parallel plates that are kept 0.15 m apart at a maximum velocity of 3 m/s. Estimate the flow rate per metre width, the shear stress on plate, the pressure difference between two points 15 m apart, the velocity gradient on the plate, and the velocity at 0.01 meters from the plates. Assume μ as 2.5 N. s/m².

8

5. (a) Derive Von-Karman Integral momentum equation for boundary layer. 8

(b) The power of a pump (P) is the function of discharge (Q), specific weight of fluid and head (H). Derive the expression of power.

7

6. (a) Discuss a constant efficiency curves for a Francis turbine. 7

(b) Design a Francis turbine runner with the following data :

Net head $H = 70$ m, speed $N = 760$ rpm, output power $P = 320$ kW, Hydraulic efficiency = 95%, overall efficiency = 87%, flow ratio = 0.16, width to outer diameter of runner ratio = 0.1, inner diameter is $1/2$ of outer diameter. Also assume 5% of circumferential area of the runner to be occupied by the thickness of the vanes. Velocity of flow remains constant throughout and flow is radial at exit. 8

7. (a) With the help of a neat sketch discuss the principle and working of a double acting reciprocating pump. 7

(b) Determine the efficiency of a four-stage centrifugal pump which is running at 500 r.p.m. to create a total head of 45 m to discharge water of $0.15 \text{ m}^3/\text{s}$. If the width and diameter of each impeller at outlet is 6 cm and 70 cm respectively having vane tip angle of 40° . Find the manometric efficiency.